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PALEOMAGNETIC RESULTS FROM THE SKAERGAARD INTRUSION, EAST GREENLAND

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The natural remanent magnetization of 22 out of a total of 31 oriented cores from the layered series of the Skaergaard gabbroic intrusion (age: 55 m.y.) in East Greenland shows good stability in thermal and AF testing. The average direction of 22 AF and 9 thermally treated specimens is $D = 170°$, $I = -59°$, $\alpha_{95} = 4.2$ before correction for tilt. The mean directions after rotation around strike to horizontal and after rotation to original attitudes suggested by others yields poorer population statistics. It is therefore concluded that flexuring took place between solidification and acquisition of remanent magnetization, a range in temperature of about 500°C which may represent an interval of somewhat less than 250,000 years. No evidence for secular variation is observed which may also suggest slow cooling through the blocking temperature range. The polarity is reversed and the pole position without "tilt correction" is 165°E, 61°N, $d_m = 6.2$, $d_p = 4.6$, which is similar to pole positions reported by others for the overlying slightly older basalt.

1. Introduction

The Skaergaard intrusion is one of an extensive series of intrusive complexes along the east coast of Greenland. It is an outstanding example of a gabbroic intrusion that underwent strong differentiation through fractional crystallization during its solidification. Detailed descriptions of its rocks and discussions of its emplacement and crystallization have been published by Wager and Deer [1] and Wager and Brown [2].

Broadly speaking, it is funnel-shaped (Fig. 1), was intruded into Precambrian gneisses and Tertiary basalts, and crystallized beneath a 2- to 3-km cover of the latter [3,4]. Brooks and Gleadow [5] have dated its emplacement at 56.4 ± 1.7 m.y. ago, somewhat later than the 55- to 60-m.y. age of the surrounding basalts. Subsequent to its emplacement a downwarping of the adjoining crust developed along the coastal flexure of East Greenland and tilted the complex towards the southeast [2].

The intrusion consists of three main structural units – a Marginal Border Group that consists of rocks that crystallized from the walls inward; an

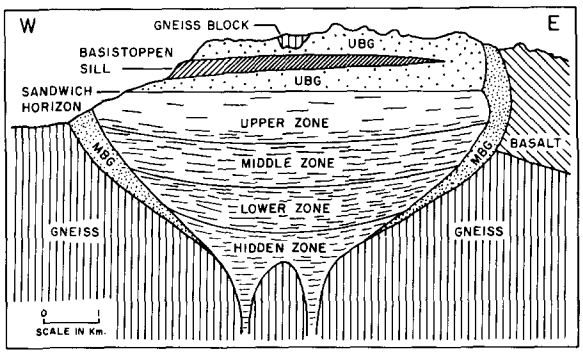


Fig. 1. Schematic west to east cross-section of the Skaergaard intrusion showing the relationships between the major units. The symbols UBG and MBG refer to the Upper Border Group and Marginal Border Group respectively; the Basistoppen Sill is one of the largest of the later intrusions (after Naslund [16]).

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Upper Border Group composed of rocks that crystallized from the roof downward; and a Layered Series, comprising the largest part of the intrusion, that consists of rocks formed from crystals that successively accumulated on the floor of the intrusion along with those that were produced by solidification of the trapped interstitial melt. Differences in the minerals present as cumulus phases in rocks of the Layered Series (see Wager and Brown [2]) permit its subdivision into a Lower Zone (LZ), a Middle Zone (MZ), and an Upper Zone (UZ), with the former and the latter being further divisible into three subunits (LZa, LZb, and LZc; and UZa, UZb, and UZc). The distribution of rocks of these various subdivisions in the area studied is shown in Fig. 2. Beneath the Lower Zone is the unexposed portion of the Layered Series, the Hidden Zone, which Wager [6] suggested represents some 70% of its volume but which McBirney [4], on the basis of more recent data, estimates as comprising slightly less than 30%. At the boundary between the top of the Layered Series and the overlapping Upper Border Group is a narrow discontinu-

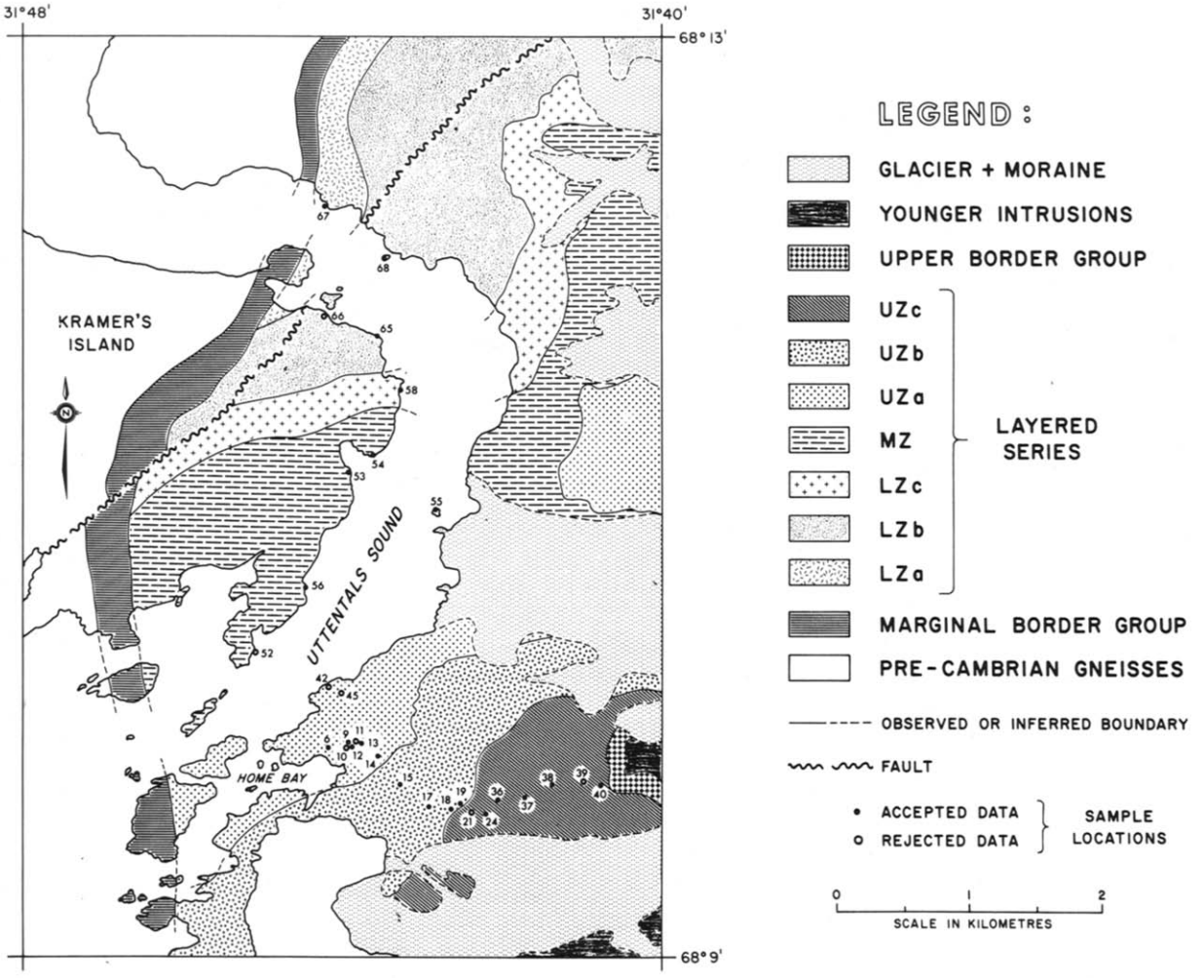


Fig. 2. Generalized geological map of the western part of the Skaergaard intrusion showing locations of specimens used in this study. Igneous layering dips generally to the southeast. The map represents a modification of that of Wager and Brown [2], resulting from largely unpublished work carried out during the summer of 1971, 1974, and 1976 by expeditions under the leadership of A.R. McBirney.

ous layer, the Sandwich Horizon, whose rocks are considered to represent products of solidification of the last Skaergaard melt. A number of gabbroic and granophyric dikes and sills cut rocks of the intrusion particularly its upper portions.

The only paleomagnetic information concerning the Skaergaard intrusion available to date were the directions of the natural remanent magnetization (NRM) determined by Girdler [7] for 14 specimens. He also determined the anisotropy of susceptibility and showed that the minimum susceptibility axes are directed perpendicular to the layering. The present study was initiated on samples collected by one of us (L.C.C.) during an expedition to Skaergaard in 1976 under the leadership of A.R. McBirney. Investigations of these specimens were carried out to obtain more precise results than those previously available and to interpret them in terms of pole position, effect of secular variation, and downwarping of the intrusion associated with the coastal flexure of East Greenland.

A JKS-10 GSC diamond drill was used to obtain core specimen $1\frac{1}{4}''$ in diameter. The orientations of the cores were measured using a Brunton compass and are considered to be accurate within 2° of azimuth of trend, and of plunge. Compass sightings of topographically distinct features were carried out at each sampling site to check the core orientations for unwanted effects due to nearby strongly magnetized rocks before the sample was accepted for paleomagnetic study. The samples were collected at irregular intervals along a traverse somewhat west of the center of the Layered Series (Fig. 2) and cover a range from about the middle of LZa to about 50 m below the top of UZc.

2. Probable cooling history

A.R. McBirney and R.M. Noyes (personal communication) suggest that the rate of upward advance of the crystallization front as the Skaergaard intrusion solidified was about 10 cm/year. In view of the work of Irvine [8], Hess [9] and Gettings [10], this appears to be a reasonable figure and, on that basis, it would have taken the approximately 2500 m of the exposed Layered Series about 250,000 years to solidify.

Williams estimates that the temperature of the Skaergaard magma at the time that the base of the Lower Zone was crystallizing, was about 1300°C. His estimate of the temperature at the time of crystallization of the top of the Upper Zone is 900°C, but McBirney [4] states that experimental evidence indicates a temperature of about 1000°C. On this basis, which if anything would correspond to a slower cooling rate, it is assumed that the exposed part of the intrusion solidified over a cooling interval of 300°C and its temperature was 1000°C when this process was just completed.

Specific heats of wall rocks, new rock and magma, and latent heat for the Skaergaard should be very similar to those of the Stillwater intrusion for which Hess [9] assumed values of 0.30–0.33 cal g^{-1} $°C^{-1}$ and 100 cal g^{-1}) respectively. On that basis, the latent heat released from the intrusion during its crystallization would be about the same as the heat released by cooling during the same interval. Accordingly, as soon as crystallization was completed, the cooling rate of the intrusion should have become about twice as rapid as it was during solidification. Thus the time required to cool the intrusion from 1000 to 500°C would be somewhat shorter than the 250,000-year period of its solidification.

3. Results of the measurements

The direction and intensity of the NRM were determined with an automated biastatic magnetometer for one or two specimens from each core. Table 1 shows that the NRM directions are to the south and up suggesting good stability of remanent magnetization and the presence of only small secondary viscous components. The NRM intensity is in the range of 10^{-3} emu cm^{-3} (1 A m^{-1}).

One specimen per core was demagnetized in stages in alternating fields of increasing peak intensity and and its magnetization measured after each stage. Table 1 shows that this treatment resulted in a marked increase in grouping up to 300 or 350 Oe peak intensity (precision parameters: k and α_{95}) and also in a significant increase in inclination. The behaviour of individual specimens is shown in Fig. 3. For most specimens, stable end points of direction were reached in rather wide ranges of alternating fields which include 300 Oe. Also, the grouping after treatment in this field is best. Consequently, the com-

TABLE 1

Average directions of remanent magnetization after various alternating field or thermal treatments *

AF(Oe)	Number of samples	D (°)	I (°)	k	α_{95} (°)	T (°C)	Number of samples	D (°)	I (°)	k	α_{95} (°)
NRM	31	165	−47	5	14.0	NRM	9	169	−52	25	10.6
50	22	168	−53	28	5.9	200	9	168	−54	27	10.1
100	22	168	−55	41	4.9	350	9	168	−55	30	9.5
200	7	168	−56	43	9.4	450	9	169	−55	31	9.4
300	22	167	−61	77	3.6	500	9	168	−55	32	9.3
350	14	171	−60	47	5.9	540	9	170	−55	29	9.7
400	9	171	−60	43	7.9	550	8	166	−56	39	9.0
500	9	172	−57	14	14.1	560	8	172	−54	20	12.7
						565	8	184	−54	10	18.7
						570	7	178	−58	18	14.4

* For conversion of units of field strength: 1 Oe (cgs) = 79.6 A m^{-1} (SI). D and I are declination and inclination respectively.

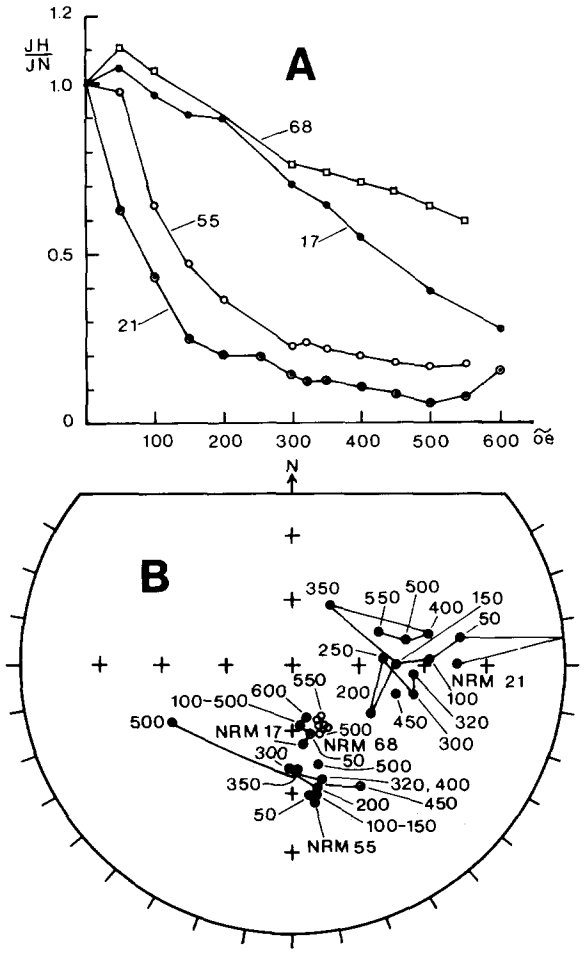


Fig. 3. Effect of alternating field demagnetization on remanent intensity (A) and direction (B).

bined results for 22 specimens were used to calculate the location of the paleomagnetic pole as discussed in a latter section of this paper. The results for 9 specimens were not accepted due either to failing to show a consistent end point (see 21 in Fig. 3) or to yielding an end point in a widely diverging direction.

Second specimens were available for 11 cores. These were used for thermal demagnetization in stages and the remanence was determined after each stage. The averaged results are shown in Table 1 while Fig. 4 shows some representative individual specimen results. In general, the results obtained after thermal treatment are comparable to those obtained after alternating field treatment but both improvement in grouping and increase in inclination are smaller.

Fig. 4A shows that the range of blocking temperature is normally narrow and that the remanence is destroyed after heating to 580°C indicating that the magnetization is carried by virtually pure magnetite. The main part of the remanence is stable against heating to at least 450°C. Fig. 4B shows that usually good end points are obtained during heating to 560°C, indicating a single component of magnetization. Best grouping is observed for 8 specimens after the 550°C heating, and these results have been used in the calculation of the paleomagnetic pole. The rest (3) of the 11 specimens showed no end point in direction (see 21 in Fig. 4) and were therefore not further considered.

Because of the small number of samples available for this study, the thermal results were combined

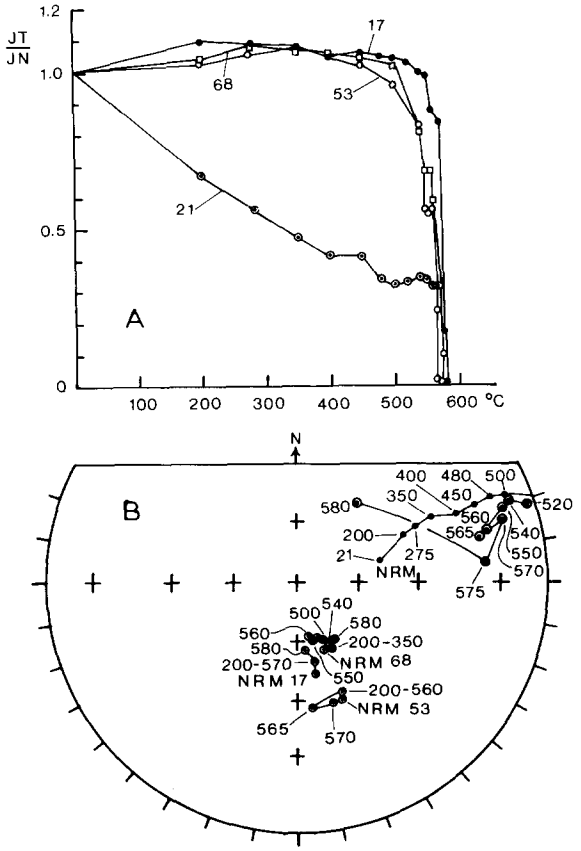


Fig. 4. Effect of thermal demagnetization on remanent intensity (A) and direction (B).

with the results obtained using alternating field demagnetization.

4. Significance of the results

4.1. Secular variation

Cooling from 1000 to 500°C of the sampled section may have taken up to about 250,000 years as discussed earlier. This suggests that cooling through the narrow blocking temperature range still took substantial time and consequently secular variation effects might be discernible in the data. For this reason, the results for the 22 specimens treated at 300 Oe and the 8 specimens heated at 550°C were combined and grouped according to the main petrographic zones from which they were taken, as shown

TABLE 2

Average directions for the three zones and of all accepted specimens (1) before any tilt correction, (2) after tilting to horizontal plane, and (3) after tilting of layering to presumed original tilt according to Wager and Brown [2]

Zone	Number of samples	D (°)	I (°)	k	α_{95} (°)
(1) Lower Zone	6	164	−68	155	5.4
Middle Zone	5	166	−55	68	8.2
Upper Zone	19	172	−57	33	5.9
All samples	30	170	−59	40	4.2
(2) Lower Zone	6	266	−83	22	14.7
Middle Zone	5	199	−68	38	10.9
Upper Zone	19	183	−76	20	7.6
All samples	30	194	−77	20	5.9
(3) Lower Zone	6	127	−76	42	10.5
Middle Zone	5	172	−67	55	9.2
Upper Zone	19	159	−76	20	7.6
All samples	30	158	−75	25	5.3

in Table 2. Evidently, there is no significant difference in average magnetic directions between these zones judging from the available samples even if the most favourable statistical population (uncorrected for tilt) is used. This suggests that either cooling through the blocking temperature range was rapid enough to occur well within one secular variation cycle (say 40°C within a few hundred years or less) or was so slow that no substantial temperature gradient existed in the intrusion while cooling through the blocking temperature range. The latter appears to be the more likely explanation.

4.2. Tilting

Appreciable flexuring, associated with the East Greenland coastal flexure, has occurred along an east-west axis in the southern part of the intrusion and the overlying basalts. Wager and Brown [2] suggest that the Skaergaard intrusion was emplaced before most of this flexuring occurred and the intrusion has been cut by dykes trending parallel to the flexure axis.

Wager and Deer [1] (see also Wager and Brown [2]) show that the attitude of layering in the intru-

sion varies between 15 and 30° to the South but their data do not show a truly systematic variation from north to south over the intrusion. By subtracting the tilt due to flexuring as suggested by the tilt of the flow planes of the overlying, older basalt, they have obtained a map with possible orientations of layering in the intrusion before substantial flexuring.

In order to compare the population statistics of the groups of directions for the three zones and all accepted samples taken together, the average directions and statistical parameters are listed in Table 2 for the following cases: (1) no correction for tilt applied, (2) after tilting to horizontal about strike, and (3) after tilting to original attitude of layering according to Wager and Deer [1]. Clearly, the grouping is best before any correction is applied to the direction of stable remanence of the individual specimens. This would suggest that flexuring occurred after solidification but before the acquisition of the stable remanent magnetization. In that case, the temperature of the intrusion was still at least 500°C at the time of deformation. A dyke contact test as suggested by Schwarz [12] would allow determination of the ambient temperature at the time of dyke intrusion and could lead to evaluation of this possibility.

The slightly older basalts exposed northeast of the Skaergaard intrusion have been the subject of several paleomagnetic studies [13,14]. The flows sampled in these studies are reported to show gentle dips. The mean directions reported in these studies after correction for tilt correspond much better to the Skaergaard average direction before correction than in the two cases also considered in Table 2. This also suggests that the uncorrected direction is a better sample of the contemporary magnetic field and the pole position derived from this direction is therefore preferable.

4.3. Pole position

The position of the magnetic pole calculated from the average magnetization direction of 31 samples uncorrected for tilt is given in Table 3, and is about 20°W of the pole position reported by Faller [14] for the oldest Tertiary basalts near the Skaergaard intrusion and about 10°W of Tarling's [13] result obtained from possibly somewhat younger basalts. The pole positions derived from the average directions of magnetization after tilting of the layering to horizontal and after tilting to the presumed original orientation of the layering [2] are also given in Table 3.

TABLE 3

Pole positions (reversed polarity) for the Skaergaard intrusion before correction for tilting of the layering, after rotating to horizontal, and after tilting to presumed original orientation. The pole positions derived for the slightly older basalt are also of reversed polarity

	Long.	Lat.	d_m	d_p
Skaergaard				
Uncorrected	165°E	61°N	6.2	4.6
Horizontal	80°E	84°N	10.9	10.2
Original	147°W	78°N	9.6	8.8
Basalt				
Faller [14]	175°W	63°N	–	–
Tarling [13]	174°E	63°N	–	–

The pole position calculated for the present layering attitudes falls at a relatively low latitude (61°). In general, Eocene poles for the North American craton are located at much higher latitudes (i.e. 85° [15]). This discrepancy in latitude is not observed if the pole position after rotating the layering attitude to horizontal is adopted. However, this would not account for the difference in dispersion listed in Table 3 and for the low latitudes of the pole positions determined by others for the basalts. The latitude of the Skaergaard pole and those of the other two poles (for the basalt) correspond well. At this time, no likely explanation for the low latitude of the pole can be selected from a number of possibilities.

References

1 R.L. Wager and W.A. Deer, Geological investigations in East Greenland, III. The petrology of the Skaergaard intrusion. Medd. Grønl. 105, 4 (1939) 1–352.
2 L.R. Wager and G.M. Brown, Layered Igneous Rocks (Oliver and Boyd, London, 1968) 572 pp.
3 D.H. Lindsley, G.M. Brown and I.D. Muir, Conditions of the ferrowollastonite-ferrohedenbergite inversion in the Skaergaard intrusion. East Greenland, in: Pyroxenes and Amphiboles: Crystal Chemistry and Phase Petrology, Mineral. Soc. Am. Spec. Publ. 2 (1969) 193–201.
4 A.R. McBirney, Differentiation of the Skaergaard intrusion, Nature 253 (1975) 691–694.

5 C.K. Brooks and A.J.W. Gleadow, A fission track age for the Skaergaard intrusion and the age of the East Greenland basalts, Geology 5 (1977) 539–540.
6 L.R. Wager, The major element variation of the layered series of the Skaergaard intrusion and a re-estimation of the average composition of the hidden layered series and of the successive residual magmas, J. Petrol. 1 (1960) 364–398.
7 R.W. Girdler, Some preliminary measurements of anisotropy of magnetic susceptibility of rocks, Geophys. J. R. Astron. Soc. 5 (1961) 197–206.
8 T.N. Irvine, Heat transfer during solidification of layered intrusions, I. Sheets and sills, Can. J. Earth Sci. 7 (1970) 1031–1061.
9 G.B. Hess, Heat and mass transport during crystallization of the Stillwater igneous complex, Geol. Soc. Am. Mem. 132 (1972) 503–520.
0 M.E. Gettings, Some thermal models of the Skaergaard intrusion, Ph.D. Dissertation, University of Oregon (1976) 171 pp.
11 R.J. Williams, Reaction constants in the system Fe-MgO-SiO_2-O_2: intensive parameters in the Skaergaard intrusion, East Greenland, Am. J. Sci. 271 (1971) 132–146.
12 E.J. Schwarz, Depth of burial from remanent magnetization: the Sudbury irruptive at the time of diabase intrusion, Can. J. Earth Sci. 14 (1977) 82–88.
13 D.H. Tarling The paleomagnetic properties of some Tertiary lavas from East Greenland, Earth Planet. Sci. Lett. 3 (1967) 81–88.
14 A.M. Faller, Paleomagnetism of the oldest Tertiary basalt in the Kangerdlugssuag area of East Greenland, Bull. Soc. Denmark 24 (1975) 173–177.
15 D.R. van Alstine and J. De Boer, A new technique for constructing apparent polar wander paths and the revised Phanerozoic path for North America, Geology 6 (1978) 137–140.
16 H.R. Naslund, Mineralogical variations in the upper part of the Skaergaard intrusion, East Greenland, Carnegie Inst. Washington Yearb. 75 (1976) 640–644.